LAB REPORT 2
K1 CSE3023 PEMBANGUNAN APLIKASI BERASASKAN WEB
SEM 4

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Exercise

Source Code

```java
/*
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 */
package com.lab.dao;

import com.lab.model.Product;
import java.sql.Connection;
import java.sql.DriverManager;
import java.sql.PreparedStatement;
import java.sql.ResultSet;
import java.sql.SQLException;
import java.util.ArrayList;
import java.util.List;

/**
 *
 * @author Asus
 */
public class ProductDAO {
    private String jdbcURL = "jdbc:mysql://localhost:3307/CSE3023";
    private String jdbcUsername = "root";
    private String jdbcPassword = "";

    // Method untuk mendapatkan sambungan Database
    protected Connection getConnection() {
        Connection connection = null;
        try {
            Class.forName("com.mysql.cj.jdbc.Driver");
            connection = DriverManager.getConnection(jdbcURL, jdbcUsername, jdbcPassword);
        } catch (SQLException | ClassNotFoundException e) {
            e.printStackTrace();
        }
        return connection;
```

```java
public class ProductDAO {

    // READ: Ambil senarai semua pengguna (Untuk View)
    public List<Product> selectAllProduct() {
        List<Product> prod = new ArrayList<>();
        String sql = "SELECT * FROM products";
        try (Connection conn = getConnection(); PreparedStatement pstmt = conn.prepareStatement(sql)) {
            ResultSet rs = pstmt.executeQuery();
            while (rs.next()) {
                int id = rs.getInt("id");

                String name = rs.getString("name");
                String category = rs.getString("category");
                Double price = rs.getDouble("price");
                int quantity = rs.getInt("quantity");
                prod.add(new Product(id, name, category, price, quantity));
            }
        } catch (SQLException e) {
            e.printStackTrace();
        }
        return prod;
    }

    // UPDATE: Kemaskini maklumat pengguna
    public boolean updateProduct(Product product) {
        boolean rowUpdated = false;
        String sql = "UPDATE product SET name = ?, category = ?, price =  ?, quantity = ? WHERE  id =  ?";
        try (Connection conn = getConnection(); PreparedStatement pstmt = conn.prepareStatement(sql)) {
            pstmt.setString(1, product.getName());
            pstmt.setString(2, product.getCategory());
            pstmt.setDouble(3, product.getPrice());
            pstmt.setInt(4, product.getQuantity());
            pstmt.setInt(5, product.getId());
            rowUpdated = pstmt.executeUpdate() > 0;
```

```java
public class ProductDAO {
    // CREATE: Masukkan pengguna baharu
    public void insertProduct(Product product) {
        String sql = "INSERT INTO products (name, category, price, quantity)VALUES( ?,  ?,  ?)";
        try (Connection conn = getConnection(); PreparedStatement pstmt = conn.prepareStatement(sql)) {
            pstmt.setString(1, product.getName());
            pstmt.setString(2, product.getCategory());
            pstmt.setDouble(3, product.getPrice());
            pstmt.setInt(4, product.getQuantity());
            pstmt.executeUpdate();
        } catch (SQLException e) {
            e.printStackTrace();
        }
    }

    // READ: Ambil satu pengguna berdasarkan ID (Untuk Update)
    public Product selectProduct(int id) {
        Product product = null;
        String sql = "SELECT id, name, category, price, quantity FROM products WHERE id =  ?";
        try (Connection conn = getConnection(); PreparedStatement pstmt = conn.prepareStatement(sql)) {
            pstmt.setInt(1, id);
            ResultSet rs = pstmt.executeQuery();
            if (rs.next()) {
                String name = rs.getString("name");
                String category = rs.getString("category");
                Double price = rs.getDouble("price");
                int quantity = rs.getInt("quantity");
                product = new Product(id, name, category, price, quantity);
            }
        } catch (SQLException e) {
            e.printStackTrace();
        }
        return product;
    }
```

```java
/**
 * Handles the HTTP <code>GET</code> method.
 *
 * @param request servlet request
 * @param response servlet response
 * @throws ServletException if a servlet-specific error occurs
 * @throws IOException if an I/O error occurs
 */
@Override
protected void doGet(HttpServletRequest request, HttpServletResponse response)
        throws ServletException, IOException {

    response.setContentType("text/html");
    PrintWriter out = response.getWriter();

    List<Product> listProduct = productDAO.selectAllProduct(); // Panggil DAO

    out.println("<h2>User List (Using DAO)</h2>");
    out.println("<table border='1'><tr><th>ID</th><th>Username</th><th>Password </th><th>Role</th><th>Actions</th></tr>");

    for (Product product : listProduct) {
        out.println("<tr>");
        out.println("<td>" + product.getId() + "</td>");
        out.println("<td>" + product.getName() + "</td>");
        out.println("<td>" + product.getCategory() + "</td>");
        out.println("<td>" + product.getPrice() + "</td>");
        out.println("<td>" + product.getQuantity() + "</td>");
        out.println("<td><a href='UpdateServlet?id=" + product.getId() + "'>Edit</a> | ");
        out.println("<a href='DeleteServlet?id=" + product.getId() + "'> Delete</a></td>");
        out.println("</tr>");
    }
    out.println("</table>");
    out.println("<br><a href='index.html'>Add New User</a>");
}
```

```java
/*
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 * Click nbfs://nbhost/SystemFileSystem/Templates/Classes/Class.java to edit this template
 */
package com.lab.model;

/**
 *
 * @author Asus
 */
public class Product {

    private int id;
    private String name;
    private String category;
    private double price;
    private int quantity;


    public Product() {
    }


    public Product(int id, String name, String category, double price, int quantity) {
        this.id = id;
        this.name = name;
        this.category = category;
        this.price = price;
        this.quantity = quantity;
    }
```
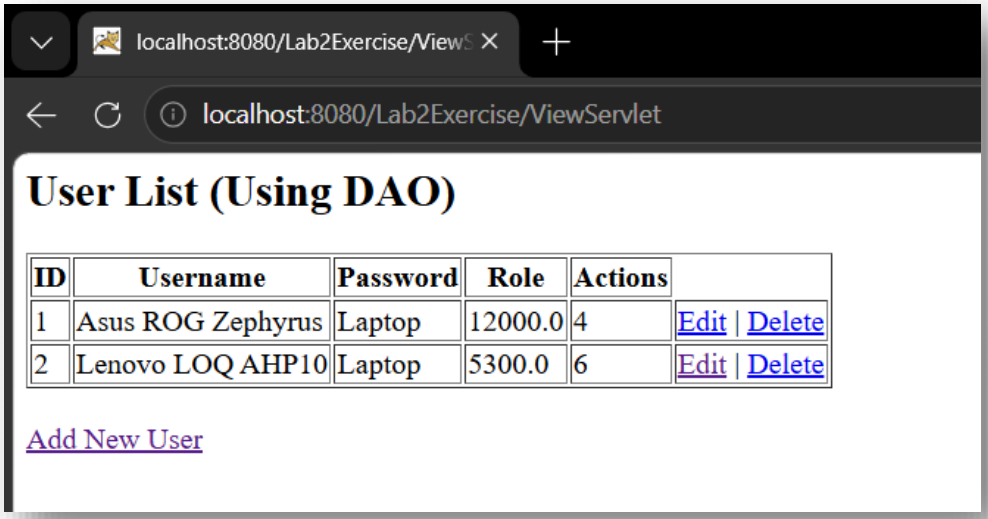

```java

    // DELETE: Padam pengguna
    public boolean deleteProduct(int id) {
        boolean rowDeleted = false;
        String sql = "DELETE FROM products WHERE id = ?";
        try (Connection conn = getConnection(); PreparedStatement pstmt = conn.prepareStatement(sql)) {
            pstmt.setInt(1, id);
            rowDeleted = pstmt.executeUpdate() > 0;
        } catch (SQLException e) {
            e.printStackTrace();
        }
        return rowDeleted;
    }
}

```

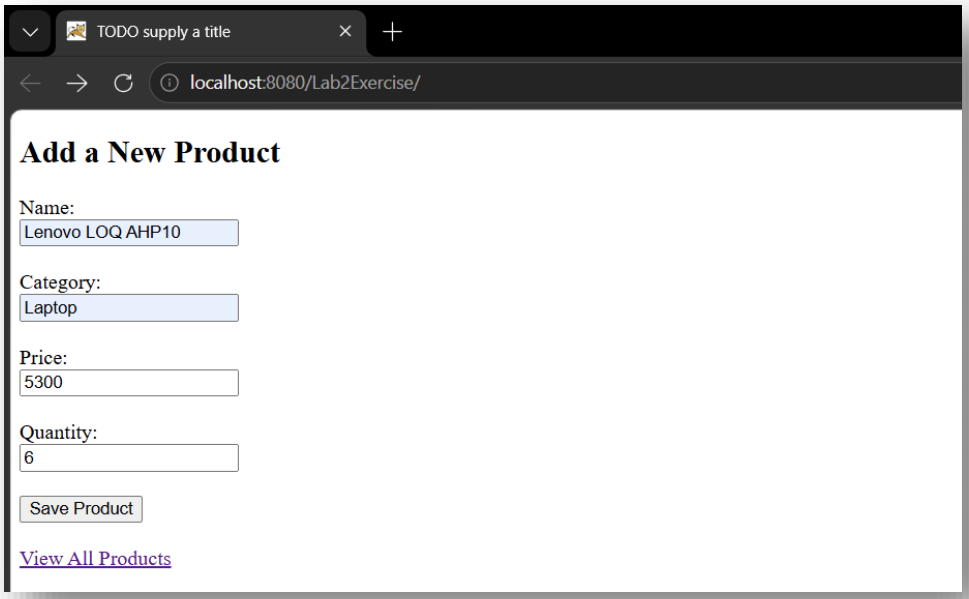
```java
public class Product {
    public Product(String name, String category, double price, int quantity) {
        this.name = name;
        this.category = category;
        this.price = price;
        this.quantity = quantity;
    }


    public int getId() {
        return id;
    }

    public void setId(int id) {
        this.id = id;
    }

    public String getName() {
        return name;
    }

    public void setName(String name) {
        this.name = name;
    }

    public String getCategory() {
        return category;
    }

    public void setCategory(String category) {
        this.category = category;
    }

```

```java
    public double getPrice() {
        return price;
    }

    public void setPrice(double price) {
        this.price = price;
    }

    public int getQuantity() {
        return quantity;
    }

    public void setQuantity(int quantity) {
        this.quantity = quantity;
    }
}
```

```java
    /**
     * Handles the HTTP <code>POST</code> method.
     *
     * @param request servlet request
     * @param response servlet response
     * @throws ServletException if a servlet-specific error occurs
     * @throws IOException if an I/O error occurs
     */
    @Override
    protected void doPost(HttpServletRequest request, HttpServletResponse response)
            throws ServletException, IOException {
        int id = Integer.parseInt(request.getParameter("id"));
        String name = request.getParameter("name");
        String category = request.getParameter("category");
        Double price = Double.parseDouble(request.getParameter("price"));
        int quantity = Integer.parseInt(request.getParameter("quantity"));

        Product product = new Product(id, name, category, price, quantity);
        productDAO.updateProduct(product);

        response.sendRedirect("ViewServlet");
    }

    /**
     * Returns a short description of the servlet.
     *
     * @return a String containing servlet description
     */
    @Override
    public String getServletInfo() {
        return "Short description";
    }// </editor-fold>

}
```

Output


localhost:8080/Lab2Exercise/ViewS
localhost:8080/Lab2Exercise/ViewServlet
User List (Using DAO)
ID | Username | Password | Role | Actions
1 | Asus ROG Zephyrus | Laptop | 12000.0 | 4 | Edit | Delete
2 | Lenovo LOQ AHP10 | Laptop | 5300.0 | 6 | Edit | Delete
Add New User


TODO supply a title
localhost:8080/Lab2Exercise/
Add a New Product
Name:
Lenovo LOQ AHP10
Category:
Laptop
Price:
5300
Quantity:
6
Save Product
View All Products

Update Product (Using DAO)

Name: Lenovo LOQ AHP10

Category: Laptop

Price: 5300.0

Quantity: 6

Update Product

localhost:8080/Lab2Exercise/ViewS

localhost:8080/Lab2Exercise/ViewServlet

User List (Using DAO)

ID	Username	Password	Role	Actions	
1	Asus ROG Zephyrus	Laptop	12000.0	4	Edit | Delete

Add New User